Formula 1 Podium Prediction Companion

Milestone 1 — Analytical report: data audit, data-engineering questions, features, models, experiments and explainability

Team members: _____

Notebook: F1_Podium_Prediction_Milestone1.ipynb (Kaggle, run-all)

Date: October 2026

Data: Ergast / Kaggle F1 World Championship 1950–2024, 14 tables

Summary

Task. For one driver entering one race, predict whether he finishes on the podium (`positionOrder ≤ 3`) using only information available **after qualifying and before the start**.

8,930	21	0.688	0.938
driver–race rows, 2003–2024, one per (raceId, driverId)	leakage-safe pre-race features in 6 groups	test PR-AUC, selected model (random forest); random = 0.15	test ROC-AUC, selected model

- **Data:** 160,494 hidden `\N` placeholders converted to real missing values; 95 columns typed with 0 failures; 85 duplicate driver-race pairs resolved; 37/37 key checks pass; raw CSVs proven unchanged by SHA-256 hashes.
- **Q1:** front-row → podium conversion rose from 68 % (2003–2013) to 76 % (2014–2024) and the circuit ranking changed substantially (ρ = 0.30). **Q2:** no home advantage once starting position is controlled for (OR 0.95, 95 % CI 0.81–1.12). **Q3:** mechanical retirements fell from 9.3 % to 6.1 % of starts and the team ranking was reshuffled by the 2022 rules (ρ = 0.28).
- **Models:** all four models beat the "grid ≤ 3" rule. The random forest had the best validation PR-AUC; on the unseen 2022–2024 test seasons it is statistically indistinguishable from the logistic-regression baseline.
- **What drives predictions:** starting position (grid + qualifying) dominates every importance measure; championship strength is second. Effects are strongly non-linear. These are associations, **not causal effects**.

1. Data audit and cleaning

All 14 tables were read as text (no automatic type guessing) and profiled with one reusable function: placeholders, primary and grain keys, foreign keys, season coverage and distributions. Raw CSVs were hashed before and after the notebook; every change happens on copies.

Issue (evidence)	Decision	Result
Missing values stored as the text `\N` — 160,494 cells in 31 columns of 6 tables; no other null-like strings	Replace with NaN on copies	NaN created = placeholders counted (asserted); 0 `\N` left
Everything is text: dates, numbers, lap and pit durations	One schema: Int64, float, explicit-format dates, timedelta, durations → seconds	95 columns, 0 conversion failures; parsed durations match `milliseconds` on 600,452 rows; memory 243 → 46 MB
85 (raceId, driverId) pairs appear 2–3 times (176 rows, 1950–1978, shared drives)	Keep the best `positionOrder`, ties → more laps → lower resultId	91 rows removed; unique grain asserted. "First row" would change 16 of 22 podium labels
139 free-text status labels	Hand-checked map into 10 groups + ambiguous flag	0 unmapped; ambiguous labels cover 2.6 % of rows
517 pit stops > 60 s, 708 laps > 5 min, 1 impossible qualifying time (1995)	Keep and flag (real red-flag events / unknown true value)	0 values changed
Entries that never started (status list or grid = 0 and laps = 0)	Removed from the modeling table; real pit-lane starters kept and flagged	1,666 rows removed, 0 podiums among them; 72 pit-lane starters kept
Qualifying missing before 1994, patchy 1996–2002 (6–59 % of races)	Model on 2003–2024	8,930 rows, 428 races, 99.8 % with a qualifying position
Standings already include the current race (jump = same-race points in 99.8 % of rows)	Use standings after the *previous* Grand Prix only	Leakage test: standings always from a strictly earlier race

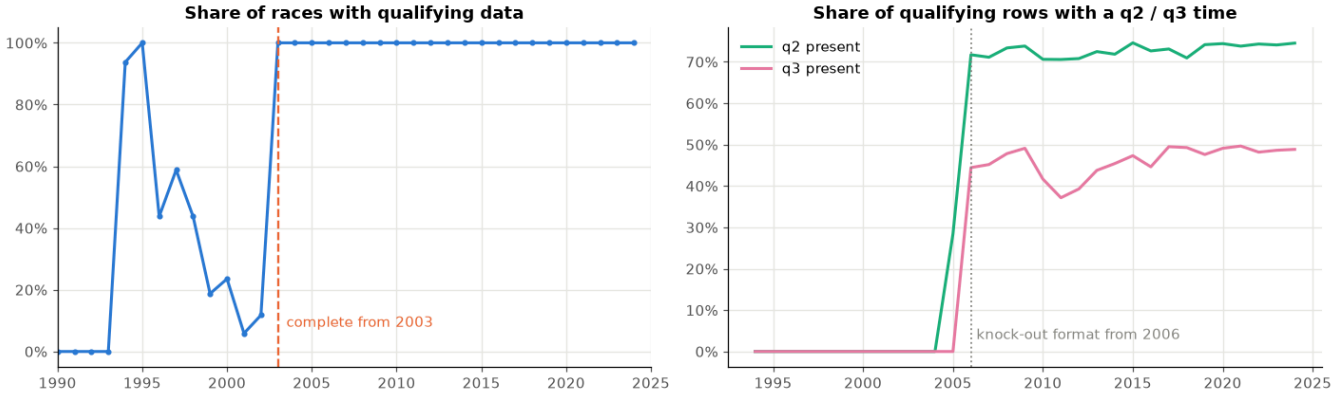


Figure 1 — Qualifying coverage by season (left) and share of rows with a Q2/Q3 time (right). Complete coverage only from 2003 → modeling period 2003–2024.

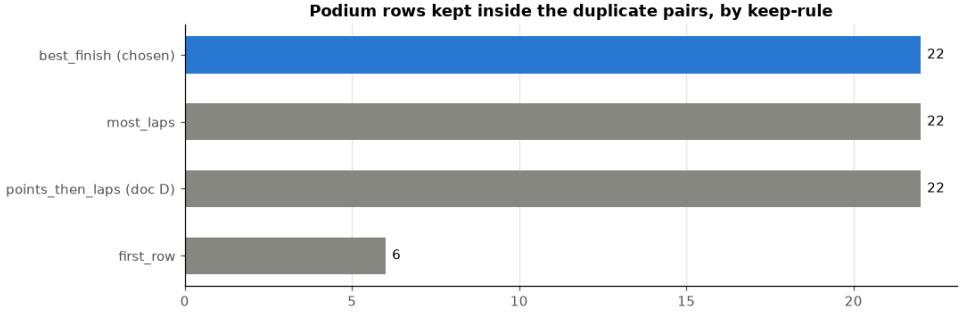


Figure 2 — Podium rows kept inside the 85 duplicate pairs by each proposed keep-rule. The chosen rule keeps every credited podium.

2. Data-engineering questions

All three questions use race **starters** only (non-starters removed) and multi-hop joins across the cleaned tables. Rates come with 95 % Wilson confidence intervals.

Q1 — Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Joins: results → races → circuits; results → qualifying. **Grain:** one row per driver start; front row = grid 1–2, and separately qualifying 1–2. **Period:** 2003–2013 vs 2014–2024 (qualifying complete in both). Circuits ranked only with ≥ 5 races in the era.

Answer. Pre-hybrid, the best converters were Magny-Cours (92 %, n = 12), Istanbul Park (79 %, n = 14) and Monza (77 %, n = 22); in the hybrid era Suzuka (89 %, n = 18), Yas Marina (86 %, n = 22) and Monaco (85 %, n = 20). **The ranking changes substantially**: rank correlation between eras ρ = 0.30, 3 of the top 5 shared, and overall conversion rose from 68 % to 76 % (dominant teams). **Grid vs qualifying:** conversion is 76 % from a front-row grid slot vs 74 % from a front-row qualifying position in the hybrid era; the 36 front-row qualifiers who started further back (penalties, pit lane) still reached the podium 31 % of the time — qualifying measures pace, grid measures track position, so both are kept as features.

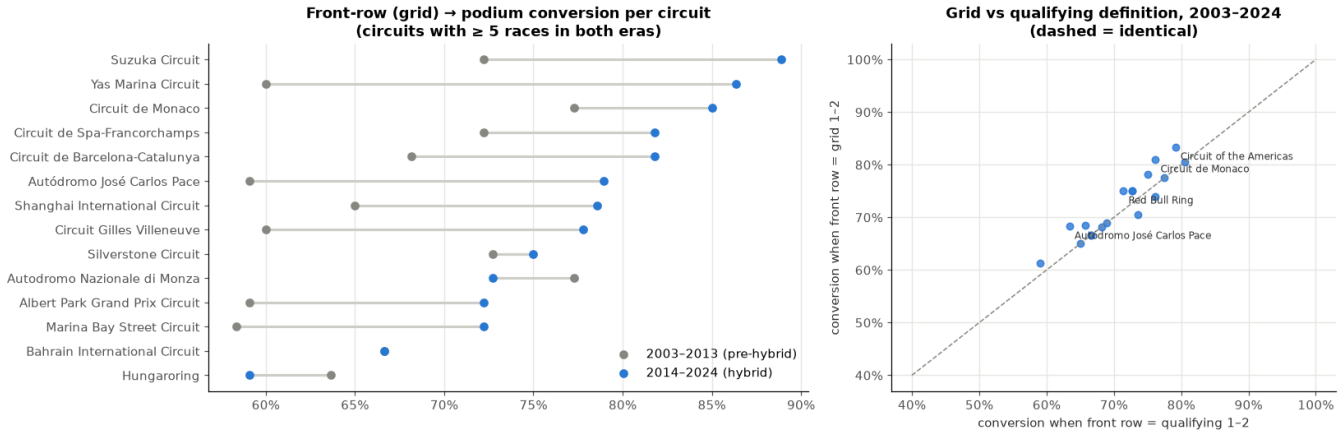


Figure 3 — Left: conversion per circuit, pre-hybrid (grey) vs hybrid (blue). Right: conversion by grid definition vs qualifying definition per circuit (dashed = identical).

Missingness: qualifying position missing for 22 starts, none on the front row. **Cutoff:** grid and qualifying are known before the start. **Caveat:** ~20 front-row starts per circuit and era → wide intervals.

Q2 — Do drivers podium more often at home, after controlling for starting position?

Joins: results → races → circuits (country); results → drivers (nationality), linked by a hand-written demonym → country map. **Control:** home vs away compared within grid bands (1–3, 4–6, 7–10, 11–15, 16+), pooled with the Cochran–Mantel–Haenszel odds ratio; robustness by logistic regression with decade controls. **Population:** starters from a grid slot, drivers with ≥ 1 home race (606 drivers, 20,806 starts, 2,313 at home).

Method	Pooled odds ratio	95 % CI	p (OR = 1)
CMH, 1950–2024 (primary)	0.95	0.81 – 1.12	0.536
Logistic regression + decade	0.92	0.78 – 1.09	0.338
CMH, 2003–2024	0.80	0.54 – 1.18	0.265
CMH, all drivers incl. no home race	0.96	0.82 – 1.13	0.649

Answer. No — there is no statistically significant home advantage once starting position is controlled for. Raw rates (home 11.5 % vs away 14.5 %) differ mainly because of where drivers start; within grid bands the differences are −1.7 to +0.5 percentage points, and every interval includes 1. The effect is the same in all bands (homogeneity p = 0.96). Consequently `home_race` was dropped in feature selection (univariate ROC-AUC 0.50).

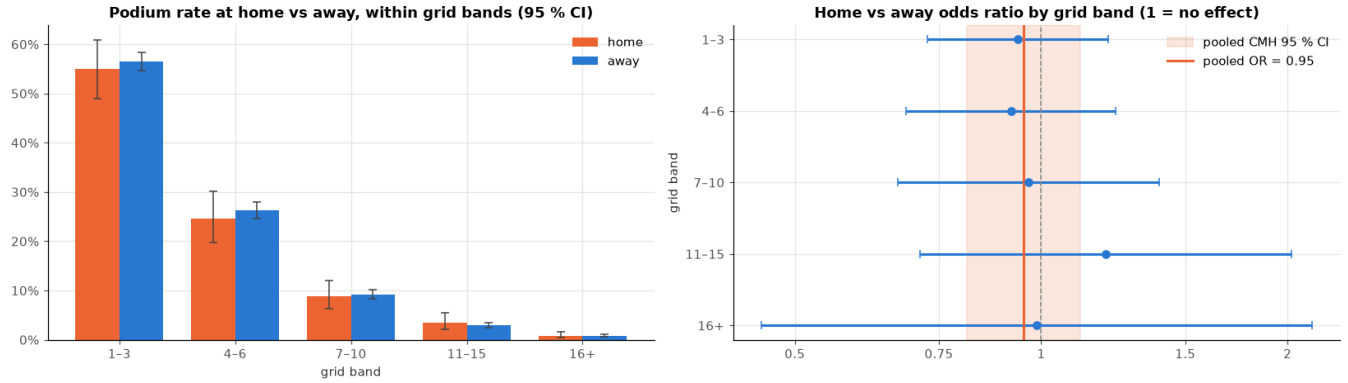


Figure 4 — Left: podium rate at home vs away within grid bands (95 % CI). Right: odds ratio per band and the pooled CMH estimate (1 = no effect).

Q3 — Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Joins: results → status (grouped) → races → constructors. **Definition:** a start that ended because a car component failed; crashes, driver causes, regulatory decisions and team errors excluded. **Rate:** mechanical retirements ÷ race starts, constructors with ≥ 40 starts. **Sensitivity:** strict (ambiguous labels such as wings, "Technical", "Stalled" removed) and broad (+ tyre/wheel failures).

Answer. 2014–2021: Lotus F1 18.7 % (14/75), Toro Rosso 16.1 % (39/242), Renault 14.0 % (28/200); Mercedes (3.8 %) and Ferrari (5.0 %) were among the most reliable. **2022–2024:** Alfa Romeo 12.5 % (11/88), Alpine 11.0 % (15/136), AlphaTauri 8.0 % (7/88); Red Bull and McLaren 3.7 %. Mechanical retirements fell from 9.3 % to 6.1 % of starts, and following re-branded teams across periods the ranking correlation is only ρ = 0.28 — the 2022 regulations reshuffled reliability. Rankings under the strict and broad definitions correlate ≥ 0.89 with the main one.

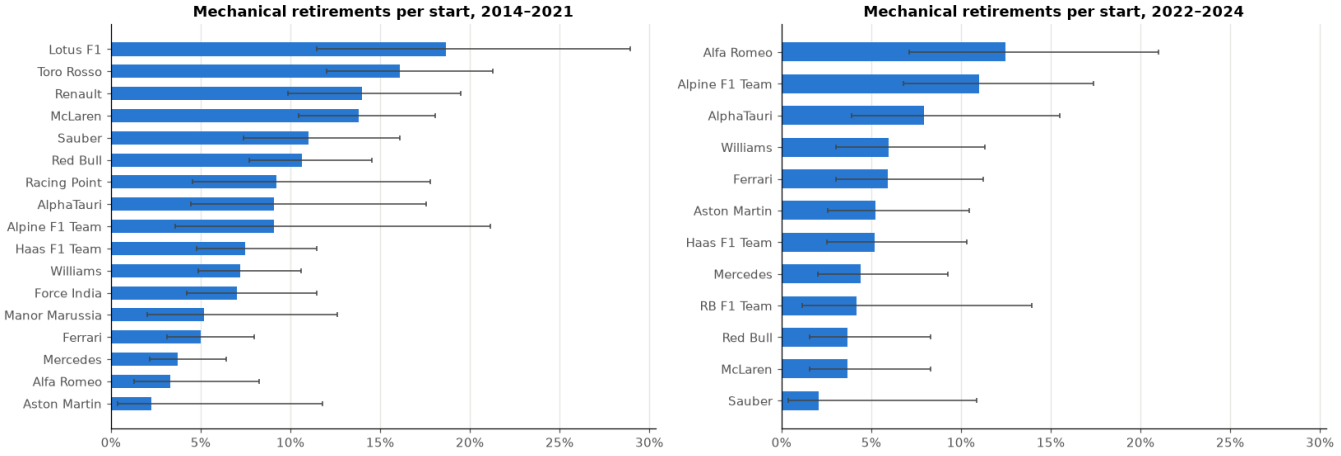


Figure 5 — Mechanical retirements per start with 95 % CI, per constructor and period.

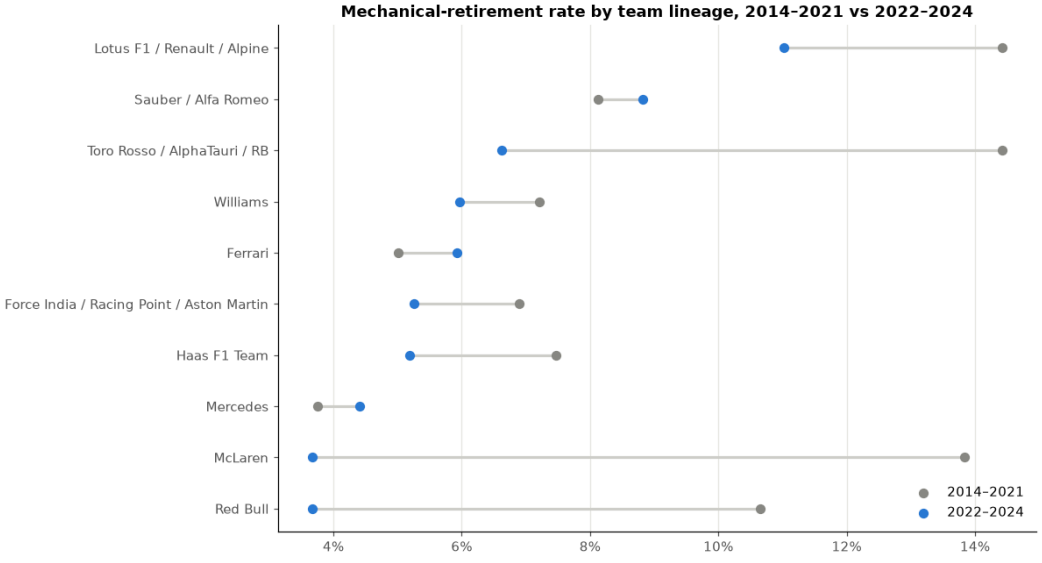


Figure 6 — The same rate following team lineages (re-brands linked) across the two periods.

Missingness: every start has a status; ~1 % carry a generic "Retired" with unknown cause. **Cutoff:** descriptive question about outcomes; used for prediction only as a lagged feature (`con_dnf_rate_last10`, past races only).

3. Features, split and preprocessing

Selected features and why each is valid at the pre-race cutoff

Group	Features	Why it is known before the start	Evidence (train ROC-AUC)
Grid	`grid`, `pit_lane_start`	Starting slot fixed after qualifying and penalties	0.91
Qualifying	`quali_position`, `gap_to_pole_pct`, `quali_top10`, `no_quali_time`, `grid_penalty`	Qualifying is finished; gap to pole is relative to the same race	0.91 / 0.89 / 0.78 / flag / 0.56
Driver form	podium rate last 5, finish % last 5, DNF rate last 10, starts before	Rolling over **previous** starts only (shifted by one; leakage-tested)	0.85 – 0.86, 0.66, 0.69
Constructor form	podiums/race last 5, best finish % last 5, DNF rate last 10	Both cars, previous races only	0.87, 0.88, 0.70
Standings	driver & constructor championship position and share of leader's points, `is_first_round`	Table after the **previous** Grand Prix; points as a share (points systems changed)	0.86 – 0.89
Context	`driver_age`, past podium rate at this circuit	Static, or previous visits only	0.58, 0.70

Selection rule fixed before validation: keep a feature if its univariate ROC-AUC on the training seasons is ≥ 0.55 (five points above chance, ~5× the sampling noise), or if it is a flag paired with a kept feature. Dropped: `home_race` (0.50, consistent with Q2) and `field_size` (0.53).

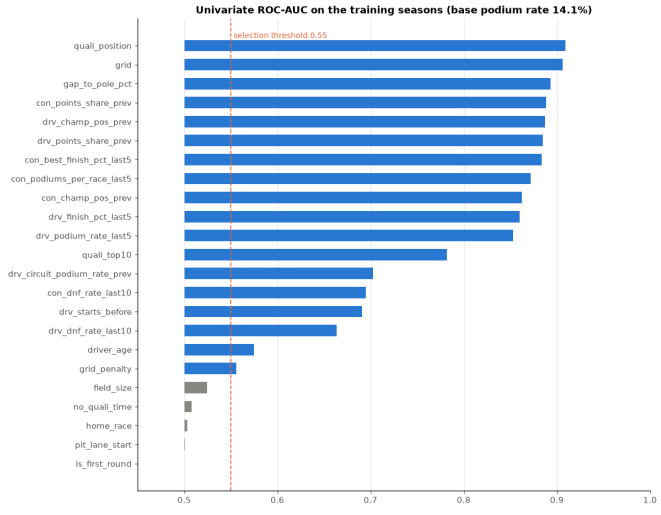


Figure 7 — Univariate ROC-AUC per candidate feature (training seasons).

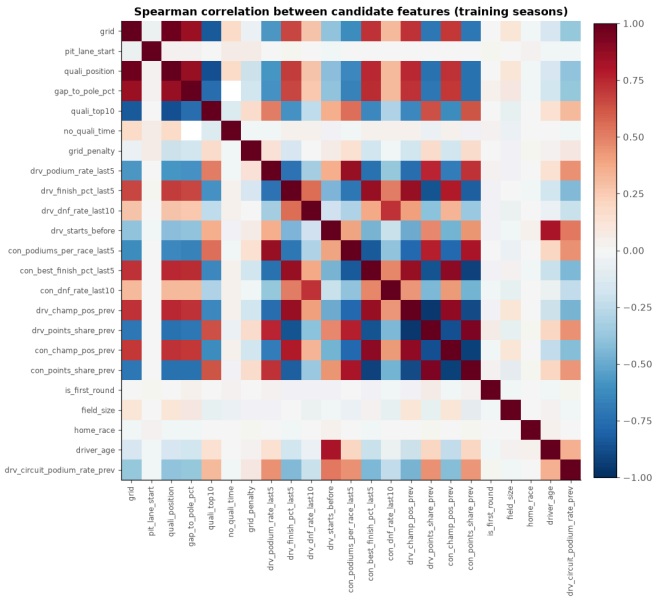


Figure 8 — Feature correlations; grid and qualifying position ρ = 0.97.

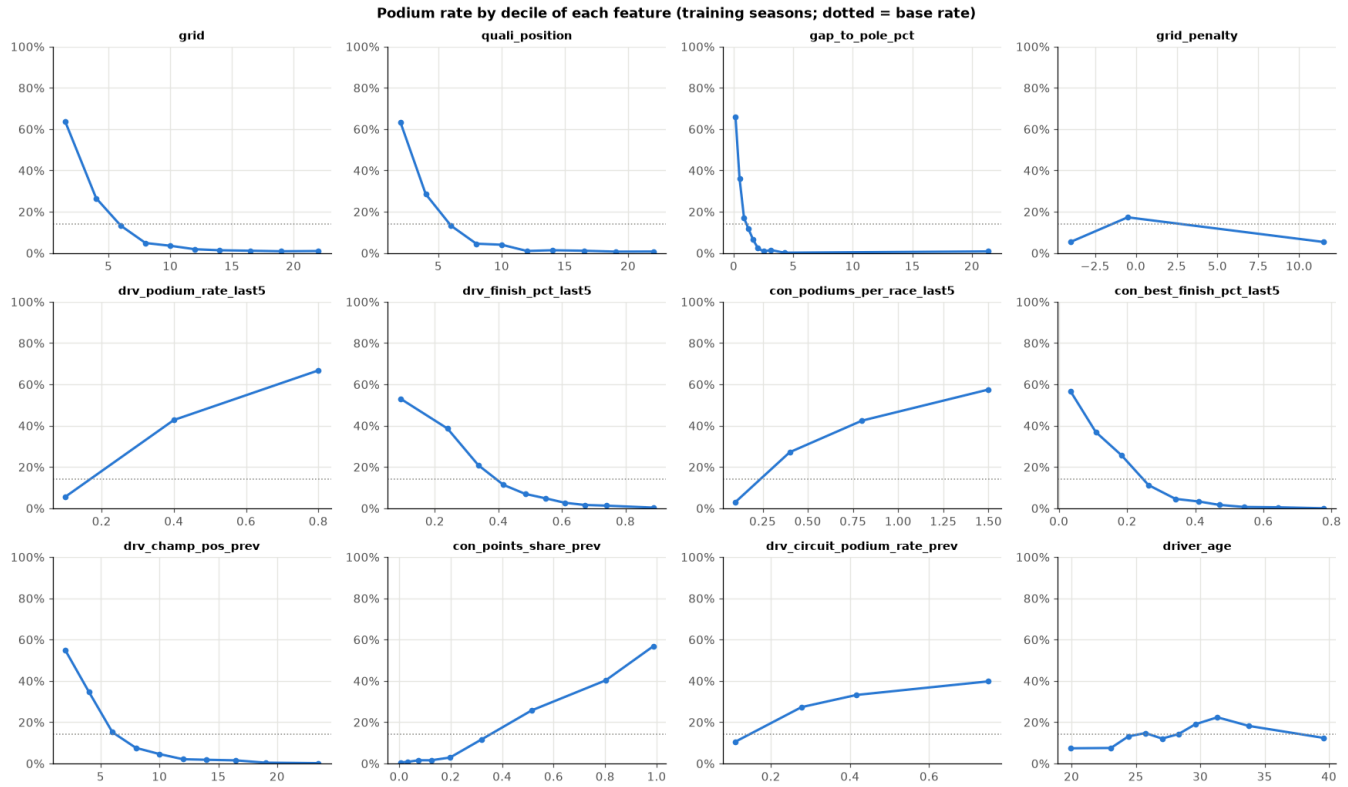


Figure 9 — Podium rate by decile of each feature (training seasons): direction and shape of each relationship. Age and grid penalty are non-linear (humps).

Train / validation / test split

Split **by season in time order**: train 2003–2018 (6,380 rows), validation 2019–2021 (1,197), test 2022–2024 (1,353). A random split would let future races inform past predictions, and the podium base rate changes with field size and era. The test seasons are the newest regulation era, the hardest and most honest test. Thresholds, hyper-parameters and preprocessing choices use validation only; the test set is scored once.

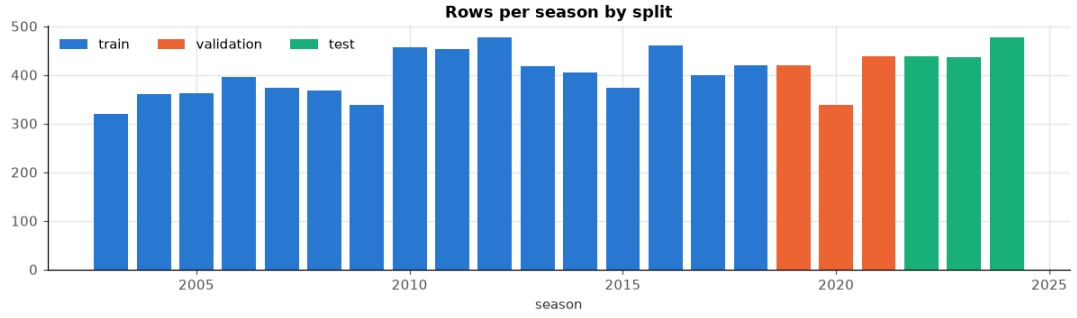


Figure 10 — Rows per season by split; podium rate 14.1 % / 15.0 % / 15.1 %.

Preprocessing and preprocessing-order experiments

Row-level F1 rules first (no qualifying position → back of the field; no time → slower than the slowest; absent from standings → behind everyone; pit lane → last grid slot), then training-median imputation with missing indicators and standard scaling, both **fit on training seasons only** inside the pipeline. Four experiments, run with the baseline and repeated with the selected model:

- **PO-1 (order):** fit imputer/scaler on all seasons vs training only — identical scores, but the first uses test statistics → rejected.
- **PO-2 (order):** impute raw qualifying times then compute the gap vs compute the gap then fill — the first produces impossible values (no-time drivers as fast as pole); with the random forest the chosen order also scores best (0.752 vs 0.740).

- **PO-3 / PO-4:** missing-value strategy and grid = 0 encoding — the rule-based options are best with the random forest in every experiment.

All differences lie inside the bootstrap noise of the validation seasons (95 % range 0.66–0.83), because the variants change only ~5 % of rows; choices are therefore made on correctness, and every result points the same way.

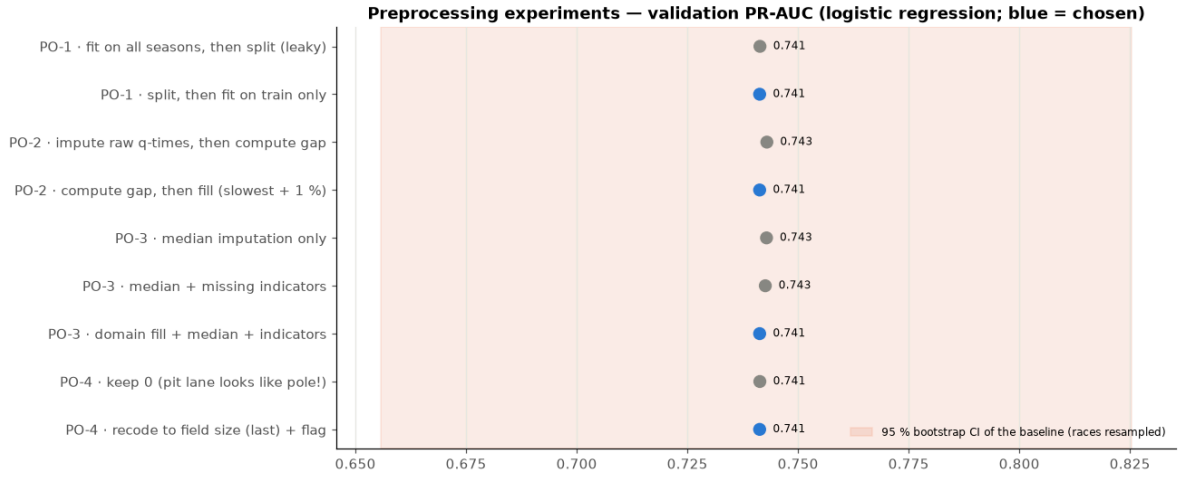


Figure 11 — Validation PR-AUC of each preprocessing variant (logistic regression) with the bootstrap noise band.

4. Predictive models

Metrics. Podiums are ~14 % of rows, so accuracy is misleading. **Primary: PR-AUC** — how well real podium finishers are ranked above everyone else (random = podium rate). **Secondary: ROC-AUC** — overall separation, comparable across eras. F1, precision and recall at the threshold that maximises F1 on validation; top-3 hit rate = share of each race's three most likely drivers who finished on the podium.

Model	PR-AUC		ROC-AUC		Test, at validation-chosen threshold			
	val	test	val	test	F1	precision	recall	top-3 hit
Rule: grid ≤ 3 (no ML)	0.677	0.647	0.917	0.914	0.649	0.650	0.647	0.652
Logistic regression (baseline)	0.743	0.702	0.939	0.936	0.680	0.567	0.848	0.647
Random forest	**0.751**	**0.688**	**0.940**	**0.938**	**0.673**	**0.616**	**0.740**	**0.657**
Gradient boosting	0.749	0.676	0.938	0.932	0.674	0.622	0.735	0.627
Neural network (MLP)	0.736	0.677	0.933	0.926	0.653	0.611	0.701	0.627

- **Logistic regression (baseline):** weighted sum of scaled features → probability; L2-regularised (C = 0.1 chosen). Limitation: linear in each feature, no curves or interactions.
- **Random forest (selected):** 400 trees on bootstrap samples, averaged (depth 6, min leaf 5). Limitation: compressed probabilities, largest train–validation gap (+0.05).
- **Gradient boosting:** shallow trees added sequentially to fix previous errors (lr 0.03, depth 3, L2 = 1). Limitation: most prone to over-fitting.
- **Shallow neural network:** one hidden layer of 32 ReLU units, Adam, L2 = 0.001, early stopping at epoch 10 on validation PR-AUC. Limitation: sensitive to initialisation, least interpretable, little room above simpler models on ~6,400 rows.

Findings. 36 configurations were tried (all kept in the notebook). The random forest had the best validation PR-AUC (0.751) and was selected; on the test seasons its PR-AUC is 0.688 (95 % CI 0.632–0.754) vs 0.702 for the baseline — a difference of −0.014 (CI −0.035 to +0.017), i.e. **not distinguishable**: the simpler baseline is an equally defensible choice. All models beat the "grid ≤ 3" rule (test PR-AUC 0.647). Test scores are below validation for every model — the expected cost of the 2022 regulation change. Validation sits above training for the linear and neural models because 2019–2021 were easier seasons to predict (the grid rule also scores higher there), not because of under-fitting.

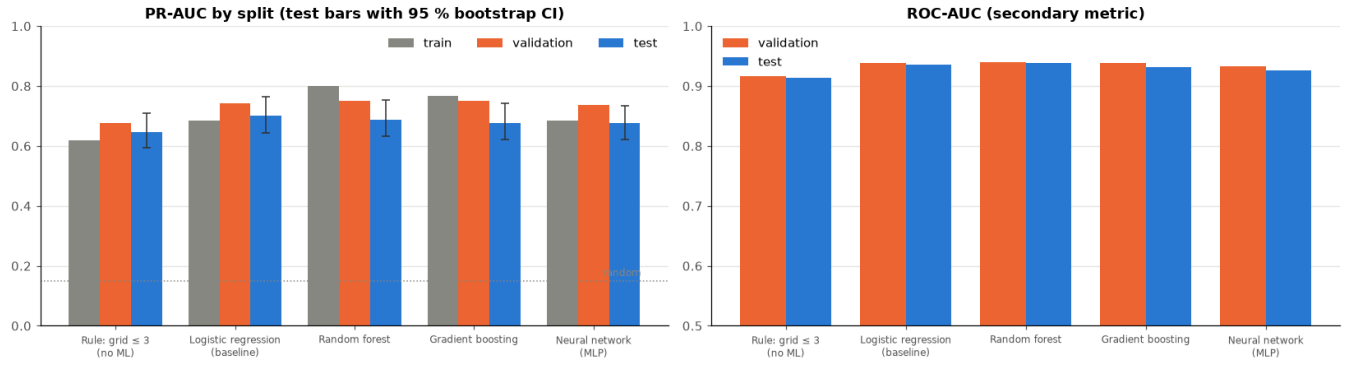


Figure 12 — PR-AUC per split (test with 95 % bootstrap CI) and ROC-AUC.

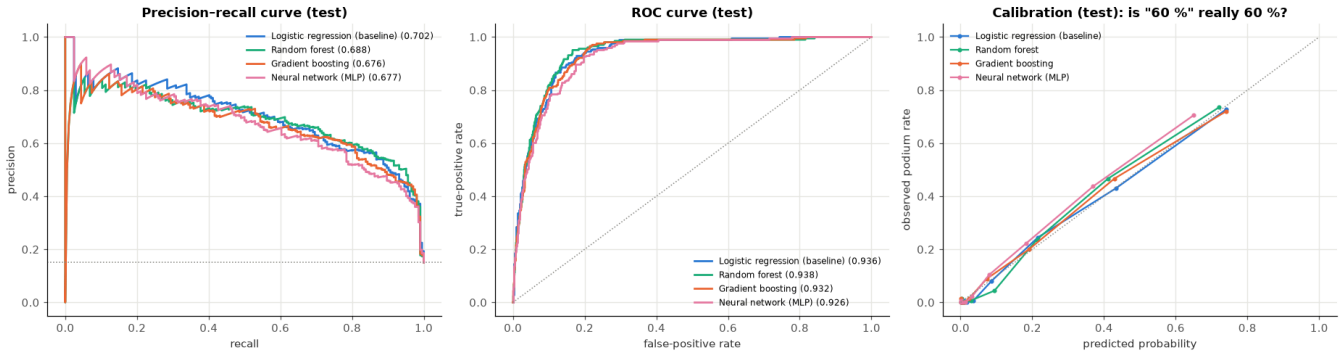


Figure 13 — Precision–recall, ROC and calibration curves on the test seasons.

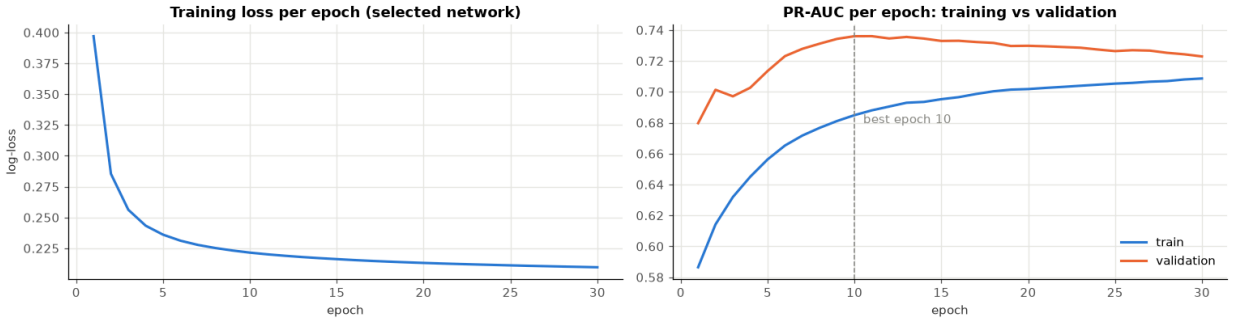


Figure 14 — Neural network: training loss and PR-AUC per epoch; early stopping at the best validation epoch.

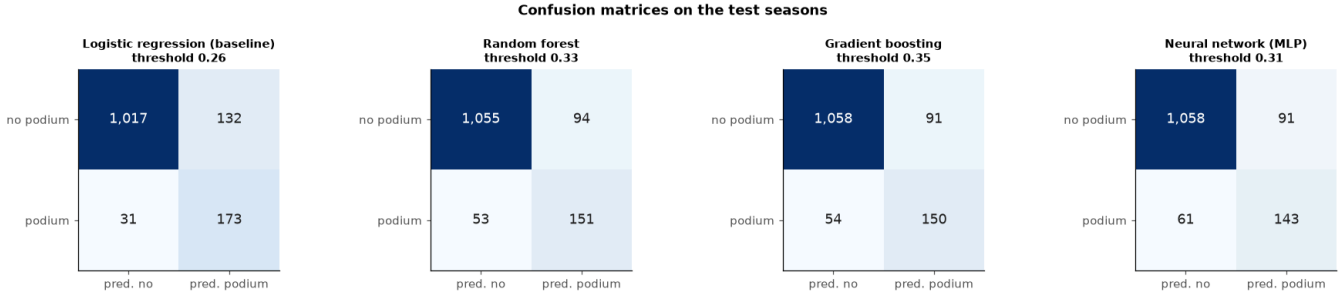


Figure 15 — Confusion matrices on the test seasons at the validation-chosen thresholds.

5. Feature-group ablation

Each feature group is removed in turn, the selected model is retrained with the same hyper-parameters, and the drop in PR-AUC is measured (95 % CI from a paired bootstrap over validation races). Groups are removed as a whole because features inside a group are strongly correlated; two combined ablations measure information shared between groups.

Removed	Features left	Validation drop	95 % CI	Test drop
grid	19	0.016	0.004 – 0.032	0.005
qualifying	16	0.023	-0.004 – 0.052	-0.017
driver_form	17	0.011	0.001 – 0.022	-0.005
constructor_form	18	0.007	0.000 – 0.013	-0.006
standings	16	0.013	-0.001 – 0.029	0.005
context	19	0.006	-0.000 – 0.013	-0.011
grid + qualifying (starting position)	14	0.111	0.047 – 0.166	0.114
standings + constructor form (championship strength)	13	0.012	-0.004 – 0.030	0.014

Reading. Every single group costs little on its own (−0.006 to −0.023), but removing **grid and qualifying together costs −0.111** (CI 0.047–0.166; −0.114 on test): the two groups carry largely the same information, so each covers for the other. Starting position is the model's dominant input; the other groups add smaller, overlapping increments.

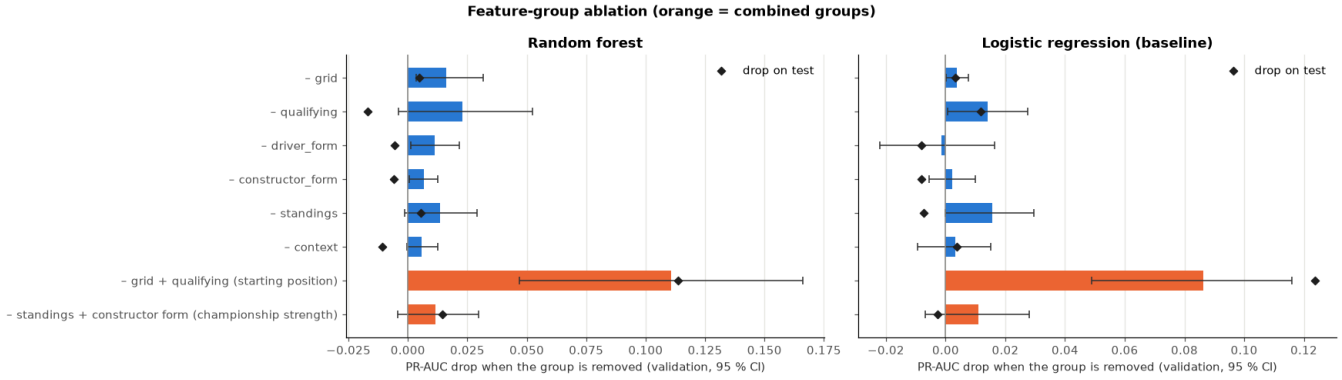


Figure 16 — Ablation for the selected model and the baseline; orange = combined groups; diamonds = drop on test.

6. Explainability (XAI)

> **Not causal.** A feature being important means the model *relies* on it to rank drivers — not that changing it would produce a podium. A strong team causes good grid slots, good standings and podiums at the same time; the explanations describe associations learned from historical data.

Permutation importance (test seasons, 20 repeats): shuffling `grid` alone costs −0.083 PR-AUC and qualifying position −0.013, but shuffling the two groups **together** costs −0.370 — per-feature importance under-states correlated features. The baseline relies on the same inputs in the same order.

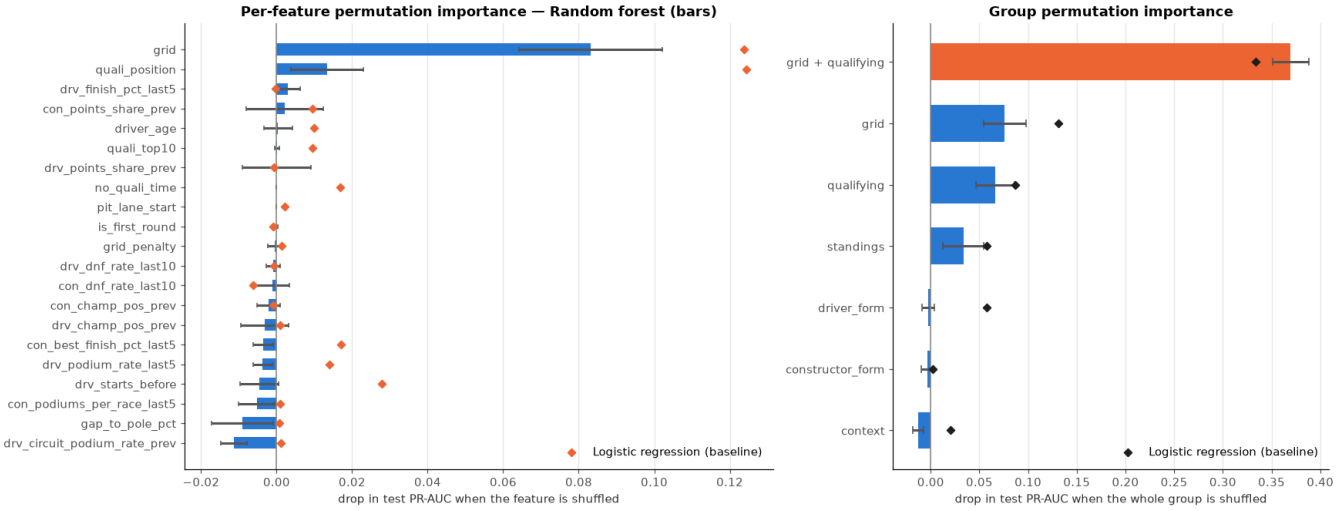


Figure 17 — Per-feature (left) and per-group (right) permutation importance; diamonds = logistic-regression baseline.

SHAP (exact TreeExplainer; additivity verified on all 1,353 test rows: base value 0.141 + contributions = predicted probability). Grid and qualifying position dominate, followed by the constructor's share of the leader's points and the gap to pole. Effects are strongly non-linear: grid 1–3 adds +0.15 to +0.25, the effect crosses zero around P5 and is flat beyond; a gap to pole above ~1 % removes most of the chance. Group rankings by SHAP agree with group permutation importance (ρ = 0.94) and with the ablation (ρ = 1.00).

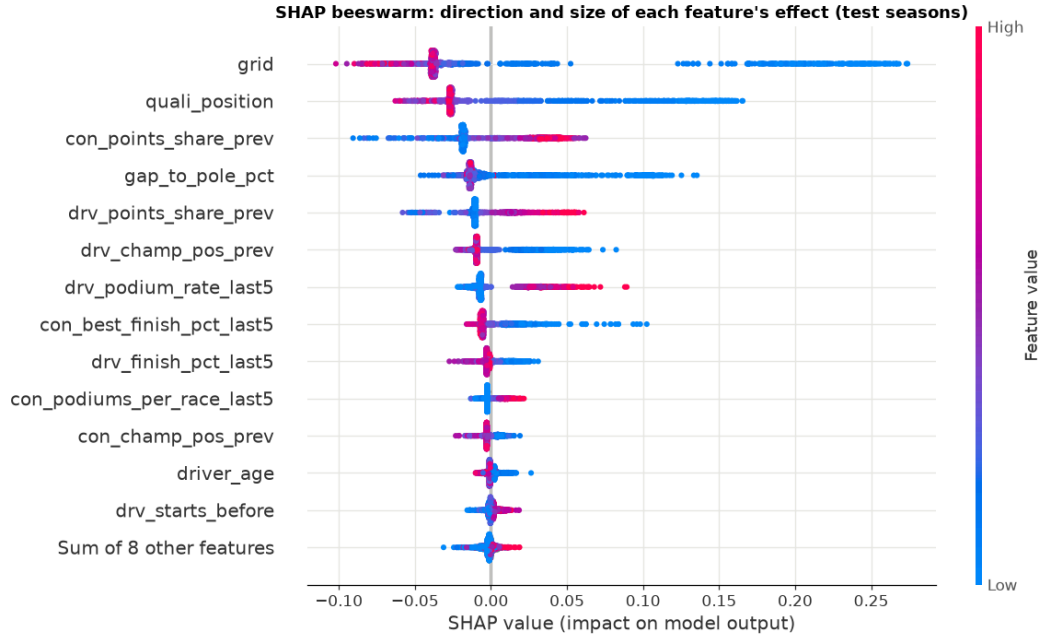


Figure 18 — SHAP beeswarm: each dot is one test driver-race; right = towards podium; colour = feature value.

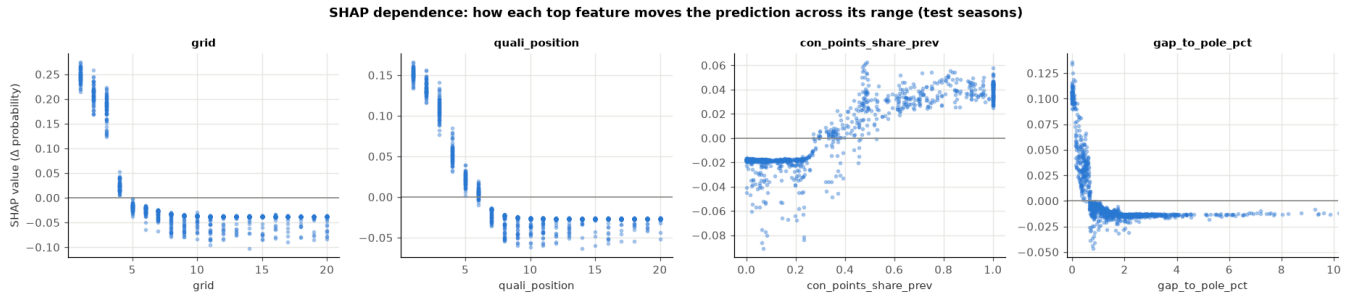


Figure 19 — SHAP dependence for the four most important features: steep near the front, flat beyond.

Local explanations

Three test cases, each explained with a SHAP waterfall and LIME. In all three, both methods agree on the top-3 features.

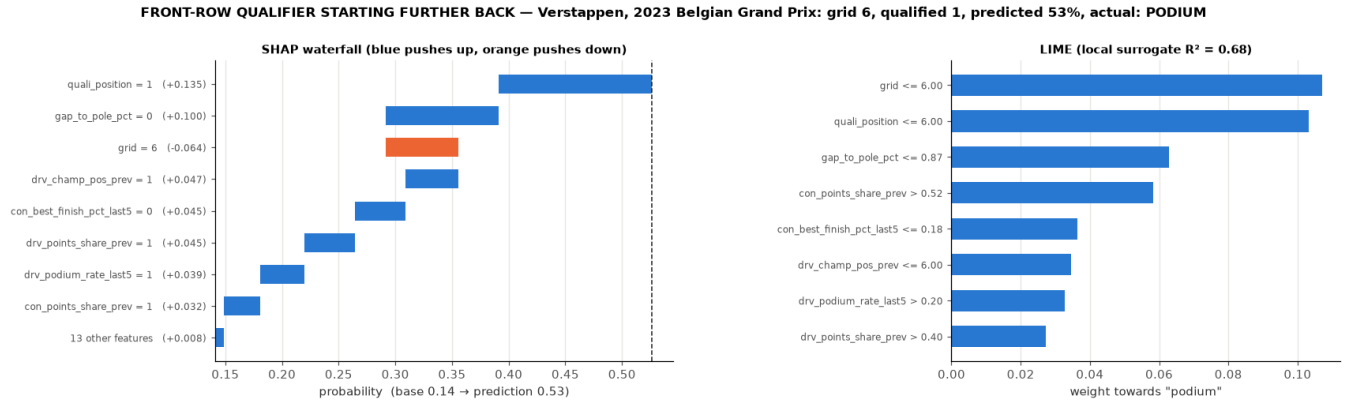


Figure 20 — Verstappen, 2023 Belgian GP: qualified 1st, started 6th after a penalty; predicted 53 %, won. Grid 6 pulls the probability down (−0.06) while qualifying pace and team strength pull it up — the Question 1 situation inside one prediction.

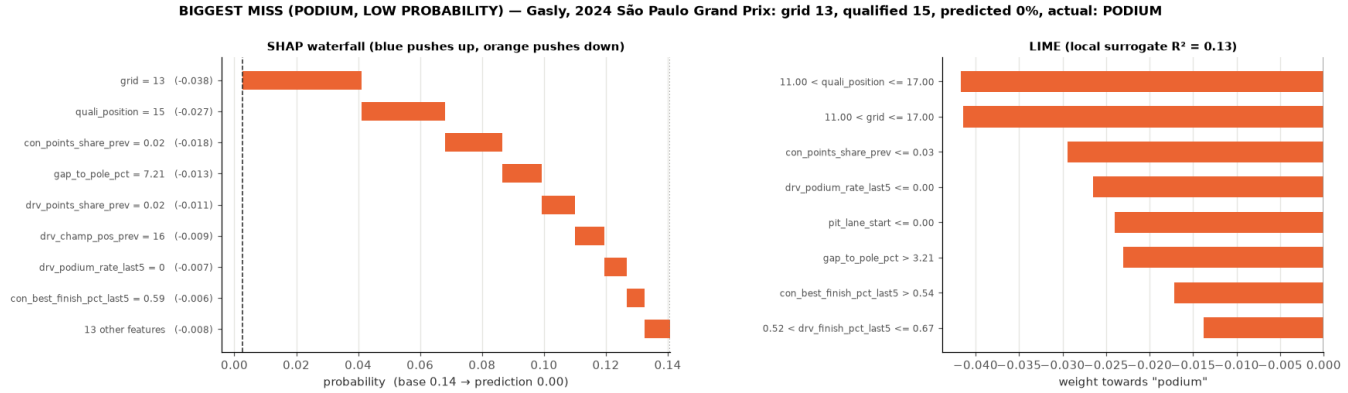


Figure 21 — Gasly, 2024 São Paulo GP: started 13th, predicted ~0 %, finished 3rd. Every pre-race input said "unlikely"; the result came from race events (a wet, chaotic race) that no pre-race model can see. LIME fits poorly here (R^2 = 0.13) — the model is non-linear around this driver.

7. Inference function

`predict_podium(record)` accepts any subset of the pre-race inputs and returns the podium probability, a decision at the validation-chosen threshold (0.333), the main local SHAP reasons and an audit of every input that was derived or filled. Inputs that are not given are treated as **unknown** (training median + missing indicator); F1 rules are applied only to known facts (no qualifying time → back of the field; pit lane → last grid slot). Out-of-range inputs are rejected.

Complete record: Verstappen, 2023 Belgian GP, all 21 inputs from data available before the race → 52.6 %, identical to the model's test prediction for that row.

Partially-missing record	Inputs given	Inputs filled	P(podium)	Podium?	Main reasons
A. only grid + qualifying known	2	20	16.5%	no	grid = 2 → +0.164; quali_position = 2 → +0.131; con_points_share_prev = unknown (median) → -0.075
B. rookie in a strong team (first start: no form, no points yet)	8	13	19.7%	no	con_points_share_prev = 0.55 → +0.060; grid = 7 → -0.035; gap_to_pole_pct = 0.60 → +0.035
C. no qualifying time, starts from the pit lane, top team	5	17	8.5%	no	grid = 20 → -0.066; con_points_share_prev = 0.90 → +0.055; quali_position = 20 → -0.050
D. same driver as demonstration 1, qualifying times lost	20	1	45.4%	yes	quali_position = 1 → +0.130; drv_champ_pos_prev = 1 → +0.059; con_best_finish_pct_last5 = 0 → +0.051

8. Limitations

- **Coverage drift:** qualifying absent before 1994 and patchy until 2002, so the model covers 2003–2024 only; lap-time, pit-stop and sprint data cover only part of the history.
- **Era sensitivity:** field size, reliability, points systems and car rules changed; test performance drops in the 2022+ regulation era, and the front-row and reliability rankings changed between eras (Q1, Q3).
- **Race-day events are invisible:** weather, safety cars and retirements ahead decide some podiums (e.g. Gasly 2024); no pre-race model can predict them.
- **Correlated features:** per-feature importance splits credit between grid and qualifying, and between standings and constructor form; group-level measures are the reliable ones.
- **Data definitions:** status labels are partly ambiguous (sensitivity checked in Q3); "home" uses the recorded nationality and the circuit's country; team form and standings reset when a team is re-branded (new constructorId); `gap_to_pole_pct` compares best times from different knock-out sessions.
- **Model choice:** the selected random forest is not significantly better than logistic regression on test; small differences between models and preprocessing variants are within the sampling noise of 3 seasons.
- **Associations, not causes:** no result in this report should be read as the effect of changing a feature.

All numbers, tables and figures in this report are produced by the run-all notebook; every decision is listed with its evidence in `decision_log.csv`.